CAMPUSCONNECT

Design Document

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Hosted application

https://campusconnect-rupam-2026.lucentterrapin1.chatgpt.site

Source repository

https://github.com/rupam1123/CampusConnect

This document describes the updated ChatGPT-hosted implementation. It supersedes the earlier MongoDB/SMTP deployment instructions in the supplied project documents.

1. Architecture

Client

React pages are compiled with Vite. Relative /api requests keep frontend and backend on the same origin. The hosted build replaces legacy password screens with verified ChatGPT sign-in and profile onboarding.

Server

hosted/worker.mjs exposes a Fetch API handler. It resolves platform identity, applies role and membership checks, and invokes shared application controllers. The compiled frontend assets are served by the same Worker.

Storage

hosted/store.mjs supplies a request-scoped D1 document adapter for the shared model/controller interface. records stores the collection name, document identifier and validated JSON. SQL indexes enforce cross-request uniqueness.

Deployment

The build produces dist/server/index.js and deployment metadata. Sites packages the exact pushed source revision, provisions DB and applies the versioned SQL migrations before running the Worker.

2. Data model

User stores the verified platform identity, email, role, name and optional student username. Professor stores the administrator-approved faculty email list.

Course references its owner, co-instructors and student roster. Session references a course and scheduled date. Comment references a session, author and optional parent comment.

Feedback references a course, session and student, with five numeric ratings. UserSessionRead stores the visitor last-seen timestamp for unread counts.

Project stores eligibility and professor ownership. Application references a project and student, with eligibility details, resume URL and review status. ForumPost and ForumComment store campus discussion content and votes.

3. Trust boundaries

Authentication headers are supplied by the hosting platform; the application does not accept user IDs or roles supplied in browser storage as authorization. The D1 binding is isolated through AsyncLocalStorage for concurrent requests.

Course membership is checked before reading discussions, sessions or unread counts. Management operations require an associated instructor; deletion requires the owning professor. Application review requires ownership of the related project.

The hosted sign-in is passwordless at the application layer. Password management belongs to the visitor ChatGPT account. There is no production email OTP delivery dependency.

Local preview uses a separate SQLite file and simulated identity for QA. That bridge is not bundled in the production Worker.